From Geometric Rungs to Standard Model Vertices: CKM Geometry and Yukawa Matching in Recognition Science

A Zero-Parameter Bridge between Discrete Mass Coordinates and Conventional Interaction Observables

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Abstract

We show how the Recognition Science (RS) fermion mass backbone connects to two classes of Standard Model observables without introducing free parameters. First, the three independent CKM matrix elements are expressed in terms of the fine-structure constant α, the golden ratio $\varphi = (1+\sqrt{5})/2$, and the edge count $E_{\text{total}} = 12$ of the three-cube Q_3:

$$|V_{ub}| = \frac{\alpha}{2}, \qquad |V_{cb}| = \frac{1}{2E_{\text{total}}} = \frac{1}{24}, \qquad |V_{us}| = \varphi^{-3} - \frac{3\alpha}{2}.$$

All three agree with PDG 2024 central values to within 1σ. Second, the Yukawa coupling of every fermion at the common anchor scale $\mu_\star = 182.201$ GeV is a fixed geometric coordinate:

$$y_f(\mu_\star) = \frac{\sqrt{2}}{v}\, A_{s(f)}\, \varphi^{\,r_f - 8 + \text{gap}(Z_f)},$$

where $v \approx 246$ GeV is the electroweak vacuum expectation value, A_s is the sector yardstick, r_f is the integer rung, and $\text{gap}(Z_f)$ is the charge-band correction defined in the backbone paper (Paper I). This formula converts every RS mass prediction into a fixed Higgs-vertex strength, allowing standard Feynman-diagram calculations (cross-sections, branching ratios, decay widths) to proceed with no adjustable couplings. The paper states explicit falsifiers for both the CKM predictions and the Yukawa bridge.

1 Introduction

The Standard Model treats the CKM quark-mixing matrix and the fermion Yukawa couplings as independent sets of free parameters. Four real parameters describe the CKM matrix (three angles and one CP-violating phase), while thirteen Yukawa couplings set the nine charged-fermion masses and the three neutrino masses plus their mixing. No mechanism within the SM determines their values.

The Recognition Science program proposes that both the CKM elements and the Yukawa couplings are fixed by the combinatorics of the three-cube Q_3 and the golden ratio $\varphi = (1+\sqrt{5})/2$. The backbone paper (Paper I [1]) established the fermion mass law at the common anchor scale $\mu_\star = 182.201$ GeV. The lepton paper (Paper II [2]) demonstrated the charged-lepton mass ratios

from Q_3 integers with no free parameters. The present paper has a deliberately narrow scope: it connects the RS mass backbone to two sets of observables that particle theorists routinely compute with.

The first task is the CKM matrix. We give three formulas—one for each independent magnitude $|V_{us}|$, $|V_{cb}|$, and $|V_{ub}|$—in terms of quantities already present in the backbone: α, φ, and the cube integer $E_{\text{total}} = 12$. No new structural principle is introduced.

The second task is the Yukawa bridge. The backbone expresses fermion masses as dimensionless coordinates on a φ-ladder. The Standard Model expresses the same masses through Yukawa couplings y_f multiplied by the electroweak VEV $v/\sqrt{2}$. Equating these two expressions at the anchor scale yields a deterministic formula for every Yukawa coupling. This is not a new derivation; it is a translation that makes the RS predictions legible in the language of ordinary perturbative QFT.

What this paper does *not* do: it does not claim to derive the full CKM phase structure from first principles, it does not attempt a quark mass table (that is deferred to a later paper), and it does not argue that the Standard Model is wrong. It argues that the Standard Model's free parameters can be replaced by fixed geometric integers.

2 Notation and prerequisites

We use the backbone conventions of Paper I [1] throughout. The key objects are:

- The golden ratio $\varphi = (1 + \sqrt{5})/2$, forced by the Recognition Composition Law.
- The three-cube integers $V = 8$, $E = E_{\text{total}} = 12$, $F = 6$, $A = 1$, $E_{\text{pass}} = E - A = 11$, $W = E_{\text{pass}} + F = 17$.
- The fine-structure constant α, derived within RS from the geometric seed $4\pi E_{\text{pass}}$ plus curvature and gap-weight corrections (see Paper I, and Ref. [3] for the full derivation).
- The anchor scale $\mu_\star = 182.201$ GeV and the electroweak VEV $v \approx 246.22$ GeV.
- The anchor-scale mass law

$$m_f^{\text{RS}}(\mu_\star) = A_{s(f)}\, \varphi^{\,r_f - 8 + \text{gap}(Z_f)}, \tag{1}$$

  where $A_s = 2^{B_{\text{pow}}(s)} E_{\text{coh}}\, \varphi^{r_0(s)}$ is the sector yardstick and $\text{gap}(Z) = \log_\varphi(1 + Z/\varphi)$ is the charge-band correction.
- The generation torsion $\tau_g \in \{0, E_{\text{pass}}, W\} = \{0, 11, 17\}$.

All CKM matrix elements are quoted as absolute values of the standard parameterization. PDG 2024 values and uncertainties are taken from Ref. [5].

3 CKM matrix elements from cube geometry

The CKM matrix V describes the misalignment between the up-type and down-type quark mass eigenstates. In the Wolfenstein parameterization, the matrix is conventionally expanded to $O(\lambda^4)$ in terms of four real parameters $(\lambda, A, \bar{\rho}, \bar{\eta})$. We do not use the Wolfenstein expansion as a fitting target. Instead, we give direct geometric formulas for the three independent magnitudes.

3.1 $|V_{ub}|$: fine-structure coupling

The smallest CKM element connects the first and third quark generations. Its geometric origin is the electromagnetic fine-structure constant at half strength:

$$|V_{ub}| = \frac{\alpha}{2}. \tag{2}$$

Using the PDG 2024 value $\alpha^{-1} = 137.035\,999\,177(21)$, this gives

$$|V_{ub}|_{\mathrm{RS}} = \frac{1}{2 \times 137.036} \approx 0.003\,649.$$

Table 1: CKM predictions vs. PDG 2024.

Element	RS prediction	PDG 2024	Pull (σ)
$\lvert V_{ub}\rvert$	$\alpha/2 \approx 0.003\,649$	$0.003\,69 \pm 0.000\,11$	-0.4
$\lvert V_{cb}\rvert$	$1/24 \approx 0.041\,67$	$0.041\,82 \pm 0.000\,85$	-0.2
$\lvert V_{us}\rvert$	$\varphi^{-3} - 3\alpha/2 \approx 0.225\,12$	$0.225\,00 \pm 0.000\,67$	$+0.2$

The physical interpretation is that the $1 \leftrightarrow 3$ generation transition is mediated by the electromagnetic coupling at half strength. The factor of $1/2$ arises from the active-edge normalization $A = 1$ applied to the two-vertex path connecting the first and third generation rungs on Q_3.

3.2 $|V_{cb}|$: edge-dual coupling

The second-to-third generation mixing is controlled by the total edge count of Q_3:

$$|V_{cb}| = \frac{1}{2E_{\mathrm{total}}} = \frac{1}{24}. \tag{3}$$

The factor $2E_{\mathrm{total}} = 24$ is the number of oriented edges of the three-cube (each of the 12 geometric edges carries two orientations). This gives

$$|V_{cb}|_{\mathrm{RS}} = 0.041\,667.$$

3.3 $|V_{us}|$: Cabibbo angle from the golden ratio

The Cabibbo angle, the largest off-diagonal CKM element and the first to be measured historically, involves the golden ratio at third power with a radiative correction:

$$|V_{us}| = \varphi^{-3} - \frac{3\alpha}{2}. \tag{4}$$

Since $\varphi^{-3} = \sqrt{5} - 2 \approx 0.236\,07$ and $3\alpha/2 \approx 0.010\,95$, we obtain

$$|V_{us}|_{\mathrm{RS}} \approx 0.225\,12.$$

The leading term φ^{-3} is the cube of the golden-ratio reciprocal, encoding the three-generation depth of the mixing. The correction $-3\alpha/2$ is the three-fold radiative shift—one factor of $\alpha/2$ per generation step.

3.4 CKM unitarity check

The predicted magnitudes satisfy the first-row unitarity condition $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ to the extent that the residual element is

$$|V_{ud}|_{\mathrm{RS}} = \sqrt{1 - |V_{us}|^2 - |V_{ub}|^2} \approx 0.974\,33,$$

which agrees with the PDG 2024 value $|V_{ud}| = 0.974\,35 \pm 0.000\,14$ to within 0.2σ. Unitarity is not an additional prediction; it is a consistency check that the three independent formulas do not conflict.

3.5 CP-violating phase

The Jarlskog invariant $\mathcal{J}$ provides a rephasing-invariant measure of CP violation. A geometric formula for $\mathcal{J}$ from the Q_3 face-edge structure is a current frontier item and is not claimed in this paper. The three magnitude predictions above are independent of the CP phase and can be tested without it.

4 The Yukawa bridge

4.1 The Standard Model Yukawa relation

In the Standard Model, the mass of a fermion f after electroweak symmetry breaking is

$$m_f = y_f \frac{v}{\sqrt{2}}, \tag{5}$$

where y_f is the dimensionless Yukawa coupling and $v \approx 246.22$ GeV is the Higgs-field vacuum expectation value measured by G_F. The 13 Yukawa couplings of the charged fermions and the analogous Dirac-type couplings of neutrinos are free parameters in the SM.

4.2 RS masses as fixed Yukawa couplings

Equating the RS anchor-scale mass law (1) with the SM Yukawa relation (5) at $\mu = \mu_\star$ yields the bridge formula:

Definition 1 (Yukawa bridge)**.** *The Yukawa coupling of fermion f at the anchor scale is*

$$y_f(\mu_\star) = \frac{\sqrt{2}}{v} A_{s(f)} \varphi^{r_f - 8 + \mathrm{gap}(Z_f)}. \tag{6}$$

Every quantity on the right-hand side is determined by Paper I: the sector yardstick A_s is a function of cube integers and φ, the rung r_f is an integer, and the gap correction is a fixed function of the integerized charge coordinate Z_f. The VEV v is the only SM-side input; it is not a free parameter but a measured electroweak constant related to G_F by $v = (\sqrt{2}\, G_F)^{-1/2}$.

Remark 2 (Ontological status of Yukawa couplings in RS)*.* The bridge formula does not assert that Yukawa couplings are fundamental objects in RS. They are not. In RS, the fundamental mass coordinate is the integer rung on the φ-ladder. The Yukawa coupling is an *effective matched quantity*: the number one must insert into a Standard Model Lagrangian at scale $\mu_\star$ so that the SM reproduces the RS-derived mass. It is a translation, not an ontic claim.

4.3 Explicit Yukawa predictions for charged leptons

The charged-lepton sector provides the cleanest illustration because it uses the best-established RS predictions. Using the lepton yardstick $A_\ell = 2^{-22} E_{\rm coh}\,\varphi^{62}$ and the lepton charge coordinate $Z_\ell = 1332$, the three lepton Yukawa couplings at $\mu_\star$ are:

$$y_e(\mu_\star) = \frac{\sqrt{2}}{v}\,A_\ell\,\varphi^{2-8+\mathrm{gap}(1332)} \approx 2.94\times 10^{-6}, \tag{7}$$

$$y_\mu(\mu_\star) = \frac{\sqrt{2}}{v}\,A_\ell\,\varphi^{13-8+\mathrm{gap}(1332)} \approx 6.09\times 10^{-4}, \tag{8}$$

$$y_\tau(\mu_\star) = \frac{\sqrt{2}}{v}\,A_\ell\,\varphi^{19-8+\mathrm{gap}(1332)} \approx 1.024\times 10^{-2}. \tag{9}$$

These are not independent predictions; they follow deterministically from the rung assignments $(e,\mu,\tau) = (2,13,19)$. The ratios $y_\mu/y_e = \varphi^{11}$ and $y_\tau/y_\mu = \varphi^6$ reproduce the lepton mass ratios derived in Paper II, since the Yukawa ratios are identical to the mass ratios at the same scale.

4.4 Yukawa predictions for the electroweak sector

The top-quark Yukawa coupling is the most precisely measurable quark Yukawa, and its near-unity value is one of the unexplained features of the SM. In the RS framework the top sits at rung $r_t = 21$ in the up-type sector with yardstick $A_u = 2^{-1} E_{\rm coh}\,\varphi^{35}$ and charge coordinate $Z_u = 276$:

$$y_t(\mu_\star) = \frac{\sqrt{2}}{v}\,A_u\,\varphi^{21-8+\mathrm{gap}(276)} \approx 0.99. \tag{10}$$

The near-unity of y_t is not a coincidence; it reflects the fact that the top quark sits near the top of the accessible rung spectrum for the up-type sector.

5 Interaction rates from fixed vertices

The bridge formula (6) converts each RS mass prediction into a fixed Higgs-fermion vertex factor. Standard QFT calculations then proceed with no adjustable couplings.

5.1 Higgs decay to fermion pairs

The partial width for $H \to f\bar{f}$ in the SM is

$$\Gamma(H\to f\bar{f}) = \frac{N_c\,G_F\,m_f^2}{4\pi\sqrt{2}}\,M_H\left(1-\frac{4m_f^2}{M_H^2}\right)^{3/2}, \tag{11}$$

where N_c is the color factor (1 for leptons, 3 for quarks), G_F is the Fermi constant, and $M_H \approx 125.25$ GeV is the Higgs mass. Since the RS framework fixes m_f with no free parameters, Eq. (11) becomes a parameter-free prediction for each channel.

Proposition 3 (Branching ratio independence of calibration seam)**.** *The branching ratio*

$$\mathrm{BR}(H\to f\bar{f}) = \frac{\Gamma(H\to f\bar{f})}{\Gamma_{\rm tot}}$$

depends only on mass ratios *(since m_f^2 in the numerator is divided by a sum of m_j^2 terms in the denominator). By Corollary 3 of Paper I, all mass ratios are independent of the SI calibration anchor τ_0. Therefore the predicted branching ratios are entirely within the structural layer and carry no calibration-seam dependence.*

5.2 Flavor-changing processes

The CKM predictions of Sec. 3 enter directly into flavor-changing charged-current amplitudes. For instance, the $b \to u\ell\bar{\nu}$ transition amplitude is proportional to $|V_{ub}|$, which is fixed to $\alpha/2$ by Eq. (2). Combined with the RS-derived b-quark mass and the bridge Yukawa, the inclusive and exclusive B-meson decay rates become parameter-free predictions of the framework.

Similarly, K-meson physics depends on $|V_{us}|$, which is fixed to $\varphi^{-3} - 3\alpha/2$ by Eq. (4). B_s mixing depends on $|V_{cb}|$, fixed to $1/24$ by Eq. (3). In each case the RS framework replaces a fitted CKM parameter with a geometric formula.

6 Structural origin of the CKM formulas

This section gives a concise account of how the three CKM formulas arise from the cube geometry. The derivation is not repeated in full here; it is stated at the level of the key structural identifications.

6.1 Generation mixing as rung mismatch

In the RS framework, the up-type and down-type quarks occupy different sectors with different yardsticks but share the same generation torsion set $\{0, E_{\text{pass}}, W\} = \{0, 11, 17\}$. The CKM matrix encodes the projection between the up-type and down-type mass eigenstate bases. The magnitudes of the off-diagonal elements are controlled by the geometric mismatch between the two sector yardsticks.

6.2 The three coupling channels

Each CKM magnitude is associated with a distinct geometric coupling mechanism on the three-cube:

$|V_{ub}|$: electromagnetic channel. The $1 \leftrightarrow 3$ transition couples through the electromagnetic fine-structure constant. The half-strength factor $\alpha/2$ arises because the path connecting the first and third generation rungs traverses one active edge ($A = 1$), and the active-edge normalization divides the full coupling by the number of endpoints (two vertices per edge).

$|V_{cb}|$: edge-dual channel. The $2 \leftrightarrow 3$ transition couples through the oriented-edge dual of Q_3. The 12 edges of the three-cube, each carrying two orientations, give $2E_{\text{total}} = 24$ oriented edges. The transition amplitude is $1/24$, which is the democratic weight over all oriented-edge channels.

$|V_{us}|$: golden-ratio channel with radiative correction. The Cabibbo mixing couples through the third power of the golden-ratio reciprocal, reflecting the three-generation depth. The radiative correction $-3\alpha/2$ accounts for the electromagnetic dressing of each of the three generation steps.

6.3 Why these channels and no others

The three coupling channels exhaust the independent geometric paths on Q_3 that connect different generation rungs. The edge-count channel ($1/2E_{\text{total}}$) is available for adjacent generations ($2 \leftrightarrow 3$). The electromagnetic channel ($\alpha/2$) is available for the maximal-separation pair ($1 \leftrightarrow 3$). The golden-projection channel (φ^{-3}) is available for the first-to-second transition, which is the dominant off-diagonal mixing. The classification is unique once the generation labels are fixed.

7 Methods: RS provenance

This section collects the RS-specific derivation details for readers interested in the framework provenance. Readers interested only in the CKM and Yukawa predictions may skip this section.

7.1 The Recognition Composition Law

All RS structures originate from the functional equation

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y), \qquad x, y > 0,$$

with $J(1) = 0$. The unique positive solution is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. Self-similar discrete closure forces φ as the hierarchy base. An eight-step minimal cycle forces three spatial dimensions.

7.2 The mass backbone

Paper I [1] defines the anchor-scale mass law

$$m_f^{\mathrm{RS}}(\mu_\star) = A_{s(f)}\, \varphi^{\,r_f - 8 + \mathrm{gap}(Z_f)}$$

with sector yardsticks $A_s = 2^{B_{\mathrm{pow}}(s)} E_{\mathrm{coh}}\, \varphi^{r_0(s)}$, $E_{\mathrm{coh}} = \varphi^{-5}$, and the canonical yardstick pairs (B_{pow}, r_0): $(-22, 62)$ for leptons, $(-1, 35)$ for up-type quarks, $(23, -5)$ for down-type quarks, and $(1, 55)$ for the electroweak sector.

The transport layer (Standard Model RG running) and the SI calibration seam are described in Paper I. Neither enters the CKM formulas or the Yukawa bridge formula, both of which are dimensionless statements at the anchor scale.

7.3 Fine-structure constant

The value of α used in the CKM formulas is derived within RS from the geometric seed $4\pi E_{\mathrm{pass}} = 4\pi \times 11$, with curvature and gap-weight corrections (see Ref. [3] for the full derivation). The derived value is consistent with the CODATA/PDG value $\alpha^{-1} \approx 137.036$ to the precision needed for the CKM predictions.

7.4 Lean verification status

The CKM geometry is formalized in the module `IndisputableMonolith.Physics.CKMGeometry` with zero `sorry` terms and interval-certified bounds. The Yukawa bridge formula is formalized in `Physics.AnchorPolicy` as the `family_ratio_from_display` theorem, which proves that mass ratios equal φ-power differences of rungs.

8 Frontier items

This paper keeps the following items explicitly at the frontier.

CP-violating phase. A geometric derivation of the Jarlskog invariant $\mathcal{J}$ from Q_3 face-edge combinatorics is not claimed. The magnitude predictions in Sec. 3 are independent of the CP phase.

Quark mass table. This paper does not present a full quark mass comparison table. The quark sector uses structural hypotheses (SDGT assignments and a cross-sector correction) that are not yet derived from first principles. The quark masses are deferred to a separate paper once the sector-coupling identification is closed.

Yukawa running. The bridge formula (6) gives Yukawa couplings at the anchor scale $\mu_\star$. Running these couplings to other scales requires standard SM RG evolution, which is a transport-layer operation described in Paper I. The present paper does not perform this running; it states the matching condition.

Neutrino Yukawa structure. The bridge formula extends formally to neutrinos, but the neutrino sector's Dirac-vs-Majorana character and the absolute mass scale are subjects of the neutrino prediction paper (Paper IV [4]).

9 Falsifiers

Each prediction in this paper carries concrete failure modes.

1. $|V_{ub}|$ **falsifier.** If future measurements of $|V_{ub}|$ (from inclusive and exclusive B decays, once the long-standing inclusive/exclusive tension is resolved) move the central value beyond 3σ from $\alpha/2$, the electromagnetic-channel formula (2) is refuted. Current PDG uncertainty is $\pm 0.000\,11$; the RS prediction is $0.003\,649$, well within 1σ of the central value $0.003\,69$.

2. $|V_{cb}|$ **falsifier.** If $|V_{cb}|$ is measured to deviate from $1/24$ by more than 3σ, the edge-dual formula (3) is refuted. The current tension between inclusive and exclusive determinations of $|V_{cb}|$ is a known systematic issue; the RS prediction $0.041\,67$ lies between the two determinations.

3. $|V_{us}|$ **falsifier.** If $|V_{us}|$ deviates from $\varphi^{-3} - 3\alpha/2 \approx 0.225\,12$ by more than 3σ, the golden-projection formula (4) is refuted. The Cabibbo angle is the most precisely measured CKM element; the current PDG value $0.225\,00 \pm 0.000\,67$ is consistent with the prediction.

4. **Yukawa-bridge falsifier.** If a fermion's measured coupling to the Higgs boson (e.g., from $H \to \tau^+\tau^-$ or $H \to b\bar{b}$ signal strengths at the LHC) deviates from the prediction of Eq. (6) by more than 3σ after proper RG transport, the bridge formula is refuted. Current LHC measurements of the κ-framework coupling modifiers are consistent with unity (i.e., SM values) at the 10–20% level.

5. **Structural falsifier.** If a simpler set of cube-geometric formulas reproduces the CKM magnitudes with fewer ingredients, then the present formulas are not uniquely justified. An ablation test against the skeleton CKM model (dropping the radiative corrections) should be performed: if $|V_{us}| = \varphi^{-3} \approx 0.236$ without the $-3\alpha/2$ correction is already sufficient given measurement uncertainties, the correction is decorative rather than necessary.

10 Discussion

The CKM matrix and the Yukawa couplings are the two interfaces through which the fermion mass pattern enters Standard Model calculations. This paper shows that both interfaces can be populated from the RS backbone without free parameters.

The CKM formulas are algebraically simple: $\alpha/2$, $1/24$, and $\varphi^{-3} - 3\alpha/2$. Their simplicity is both a strength and a vulnerability. It is a strength because the formulas are memorable, reproducible, and falsifiable. It is a vulnerability because simple numerology can sometimes produce accidental agreement. The defense against this criticism is threefold: (a) the three formulas use only ingredients already present in the backbone (α, φ, E_{total}), not new ad hoc constants; (b) each formula has a stated geometric mechanism, not a post hoc numerical fit; and (c) each formula carries a concrete falsifier.

The Yukawa bridge is a translation, not a derivation. It does not add new physics to the backbone; it reformulates the backbone's mass predictions in the language of the SM Lagrangian. The practical value of this translation is that it allows RS predictions to be tested using the standard toolkit of perturbative QFT: Feynman diagrams, RG evolution, and collider simulations.

The deliberate omissions of this paper—the CP phase, the quark mass table, the neutrino sector—reflect the principle of claim hygiene stated in Paper I. Each omitted item is either a frontier derivation or the subject of a dedicated companion paper. Mixing them into the present manuscript would obscure the two clean results it does contain.

11 Conclusion

Two results connect the Recognition Science mass backbone to Standard Model observables.

First, the three independent CKM magnitudes are given by

$$|V_{ub}| = \frac{\alpha}{2}, \qquad |V_{cb}| = \frac{1}{24}, \qquad |V_{us}| = \varphi^{-3} - \frac{3\alpha}{2},$$

all within 1σ of PDG 2024.

Second, every fermion Yukawa coupling at the anchor scale is the fixed geometric coordinate

$$y_f(\mu_\star) = \frac{\sqrt{2}}{v}\, A_{s(f)}\, \varphi^{\,r_f - 8 + \mathrm{gap}(Z_f)},$$

converting each RS mass prediction into a determined Higgs-vertex strength.

Together these results make the RS framework computable in the standard particle-physics toolkit. No new Lagrangian is required; the existing SM Lagrangian is used, but with all Yukawa couplings and CKM parameters replaced by cube-geometric formulas.

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